FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

OPEN TEST

PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	B	24.	C	47.	C
2.	B	25.	D	48.	A
3.	B	26.	B	49.	D
4.	A, C	27.	A, C, D	50.	A, B, C
5.	A, B, C, D	28.	A, D	51.	A, B, C, D
6.	B, C	29.	A, B, C	52.	A, B
7.	B, C, D	30.	A, B, D	53.	A, B, C
8.	A, B, C	31.	A, B, D	54.	A, C
9.	D	32.	C	55.	B
10.	C	33.	D	56.	A
11.	8	34.	4	57.	3
12.	6	35.	4	58.	1
13.	9	36.	2	59.	4
14.	6	37.	9	60.	1
15.	3	38.	7	61.	6
16.	3	39.	3	62.	7
17.	4	40.	1	63.	1
18.	2	41.	8	64.	1
19.	5	42.	4	65.	1
20.	4	43.	2	66.	8
21.	02200.00	44.	00252.65	67.	00000.25
22.	00004.37	45.	01299.00	68.	00000.20
23.	00009.37	46.	00123.00	69.	00003.50

Physics

PART – I

SECTION – A

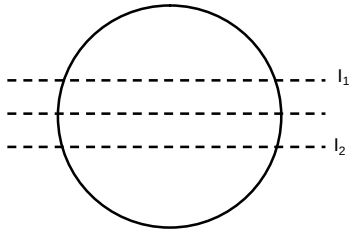
1. $\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{v}+\frac{1}{-20}=\frac{1}{-40} \Rightarrow v=+40\,mm$

$m=\frac{-v}{u}=-\left(\frac{+40}{-20}\right)=+2$

$\Rightarrow$ separation between the two images formed = $\sqrt{(20)^2+(1)^2} \approx 20.025$ mm

2. Path difference at angular position θ is d sin θ.

3. Let $\rho\frac{V}{2}=m$



$r_{CM}=\frac{m\left(R+\frac{3R}{8}\right)+2m\left(R-\frac{3R}{8}\right)}{3m}=\frac{7R}{8}$

Moment of Inertia of sphere about a horizontal are passing through the hinge.

$I=\left[I_1+m\left(R+\frac{3R}{8}\right)^2\right]+\left[I_2+2m\left(R-\frac{3R}{8}\right)^2\right]$

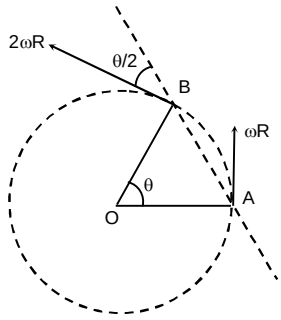
$I=\left[\frac{2}{5}mR^2-m\left(\frac{3R}{8}\right)^2+m\left(\frac{11R}{8}\right)^2\right]+\left[\frac{2}{5}(2m)R^2-2m\left(\frac{3R}{8}\right)^2+2m\left(\frac{5R}{8}\right)^2\right]=\frac{69mR^2}{20}$

$\tau=6mgR\theta-3mg\times\frac{7R}{8}\theta=\frac{27}{8}mgR\theta$

$I\frac{d^2\theta}{dt^2}+\frac{27}{8}mgR\theta=0$

$\Rightarrow \quad T=2\pi\sqrt{\frac{46R}{45g}}$

4. $f'=f_0\left[\frac{1-\frac{1}{50}\cos\frac{\theta}{2}}{1-\frac{1}{100}\cos\frac{\theta}{2}}\right]$, $\theta=\omega t$



$k=\frac{\omega R}{v}=\frac{1}{100}$

$t=\left[\frac{1-2k\cos\left(\frac{\theta}{2}\right)}{1-k\cos\left(\frac{\theta}{2}\right)}\right]$

$\frac{dt}{d\theta}=\frac{\left[1-k\cos\left(\frac{\theta}{2}\right)\right]\left[+k\sin\left(\frac{\theta}{2}\right)\right]-\left[1-2k\cos\left(\frac{\theta}{2}\right)\right]\left[\frac{k}{2}\sin\left(\frac{\theta}{2}\right)\right]}{\left[1-k\cos\left(\frac{\theta}{2}\right)\right]^2}$

$\Rightarrow \quad \frac{k}{2}\sin\left(\frac{\theta}{2}\right)=0 \Rightarrow \frac{\theta}{2}=0,\pi \Rightarrow \theta=0,\ 2\pi$

$$f_{min} = \frac{98}{99} f_0 \text{ at } \theta = 0°$$

$$f_{max} = \frac{102}{101} f_0 \text{ at } \theta = 2\pi$$

5. Basic knowledge to write angular momentum and kinetic energy of the system.

6. $x = \frac{y}{2}$, and $F(y) = \frac{mgx}{\ell} = \frac{mgy}{2\ell}$

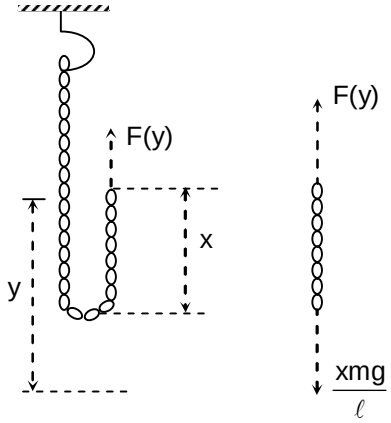
Work done by $F(y) = \int_0^{\ell} F(y)dy = \frac{mg}{2\ell}\int_0^{\ell} ydy = \frac{mg\ell}{4}$

Second method: Using WET, we can write

$K_f - K_i = W_F + W_{mg} = 0$

$\Rightarrow \quad W_F = -W_{mg} = -\left[-mg\left(\frac{\ell}{2} - \frac{\ell}{4}\right)\right] = \frac{mg\ell}{4}$

Concept: Work energy power



7. For resonance

$I_{20} = I_{10}\sin\theta$

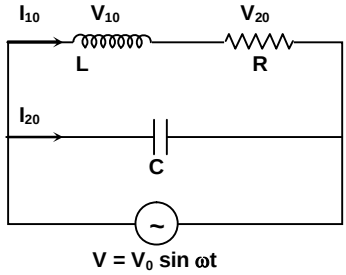
$\frac{V_0}{X_c} = \frac{V_0}{\sqrt{X_L^2 + R^2}} \times \frac{X_L}{\sqrt{X_L^2 + R^2}}$

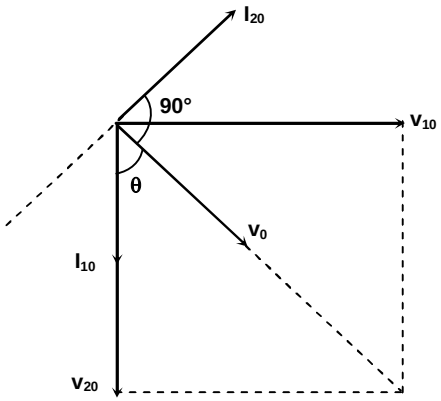
$\Rightarrow \quad X_L^2 + R^2 = X_C X_L$

$\Rightarrow \quad L^2\omega^2 + R^2 = \frac{L}{C}$

$\Rightarrow \quad \omega^2 = \frac{1}{LC} - \left(\frac{R}{L}\right)^2$

$\Rightarrow \quad \omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{L}\right)^2}$





8. By energy conservation

$\frac{1}{2}\frac{Mm}{m+M}v^2 = \frac{1}{2}kx^2$

$\Rightarrow \; x = J\sqrt{\frac{m}{M(M+m)K}} = 5\sqrt{\frac{1}{2\times 3\times 6}} = \frac{5}{6}m$

9. $v^2 = \frac{GM}{R^3}(R^2 - x^2)$

10. The waves emitted from the source cease to reach the recorder after the source's speed exceeds that of sound:

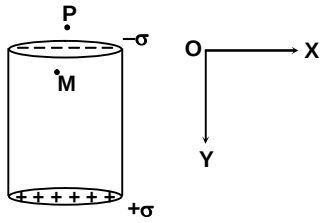
$v_0 = \omega R \sin \omega t;$

Where $\omega = \sqrt{\frac{g}{R}}$

The corresponding x = R cos ωt = $R\sqrt{1 - \frac{v_0^2}{\omega^2}}$

SECTION – C

11. Assume that on circular face of cylinder, a charge q of surface charge density σ(in magnitude) appears. So,



Q = sσ, $\vec{E}_P = \frac{\sigma}{\varepsilon_0}\hat{j}$, $\vec{E}_{M(Q^-)} + \vec{E}_{M(q)} + \vec{E}_{M(Q^+)} = 0$ …(1)

$\vec{E}_{P(q)} = \frac{q}{4\pi\varepsilon_0 a^2}(+\hat{j})$

$\vec{E}_{P(Q^-)} + \vec{E}_{P(q)} + \vec{E}_{P(Q^+)} = \frac{\sigma}{\varepsilon_0}\hat{j}$ …(2)

From equation (1) and (2)

$\vec{E}_{P(q)} = \frac{\sigma}{\varepsilon_0}\hat{j}$

$\Rightarrow \quad \sigma = \frac{q}{4\pi a^2} \Rightarrow Q = \frac{qS}{4\pi a^2}$

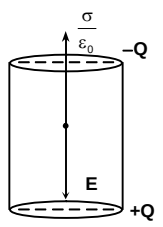
$\left|\vec{F}_{cylinder}\right| = \frac{qQ}{4\pi\varepsilon_0}\left[\frac{1}{a^2} - \frac{1}{(a+\ell)^2}\right] = \frac{qQ}{4\pi\varepsilon_0 a^2(a+\ell)^2}\left[(a+\ell)^2 - a^2\right]$

$= \frac{qQ(\ell^2 + 2a\ell)}{4\pi\varepsilon_0 a^2(a+\ell)^2} \approx \frac{2qQa\ell}{4\pi\varepsilon_0 a^4}$

$= \frac{q\ell}{2\pi\varepsilon_0 a^3} \times \frac{qS}{4\pi a^2} = \frac{q^2}{8\pi^2\varepsilon_0 a^5} V$

Second Method:

Due to induction, $-Q = -s\sigma$ and + Q = + σs will appear at upper surface and lower surface of conductor. These two charged system will act like a capacitor for region inside the conductor, so due these charges, net electrostatic field at inside the conductor has magnitude $\frac{\sigma}{\varepsilon_0}$

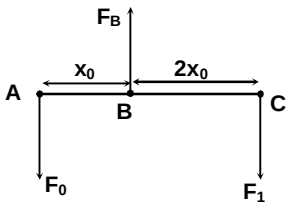


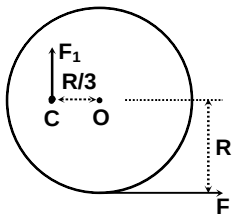
Since net field due to all charges inside the conductor is zero.

$E = \frac{\sigma}{\varepsilon_0} \Rightarrow \frac{q}{4\pi\varepsilon_0 a^2} = \frac{\sigma}{\varepsilon_0}$

$\Rightarrow \quad \sigma = \frac{q}{4\pi a^2}$

12. $F_1 = 3F$ and $F_0 = 2F_1 = 6F = 60$ N

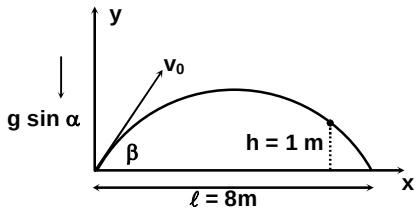




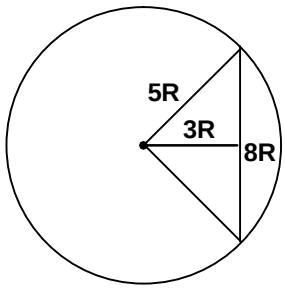
13. $h = \ell\tan\beta - \dfrac{1}{2}\dfrac{g\sin\alpha\ell^2}{v_0^2\cos^2\beta}$

$v_0 = \sqrt{\dfrac{g\sin\alpha\ell^2}{2\cos^2\beta(\ell\tan\beta - h)}}$

$= \sqrt{\dfrac{10\times\frac{4}{5}\times 8\times 8}{2\times\frac{4}{5}\times\frac{4}{5}\left(8\times\frac{3}{4}-1\right)}} = 9$ m/s

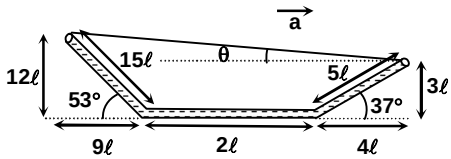


14. $\phi = \dfrac{q_{in}}{\varepsilon_0} = \dfrac{\rho\left(\frac{1}{2}8R\cdot 3R\right)R}{\varepsilon_0} = \dfrac{12\rho R^3}{\varepsilon_0}$



15. $\dfrac{9\ell}{15\ell} = \dfrac{3}{5} = \dfrac{a}{g}$

$a = \dfrac{3}{5}g$



16. $T = 2\pi\sqrt{\dfrac{I}{mgd}}$

$f = \dfrac{1}{2\pi}\sqrt{\dfrac{mgd}{I}}$

$f_0 = \dfrac{1}{2\pi}\sqrt{\dfrac{(2m)(g)(3\ell/4)}{\left(\frac{m\ell^2}{3}+m\ell^2\right)}} = \dfrac{1}{2\pi}\sqrt{\dfrac{18g}{13\ell}}$

When disc is removed

$f = \dfrac{1}{2\pi\sqrt{\dfrac{mg(\ell/2)}{m\ell^2/12}}} = \dfrac{1}{2\pi}\sqrt{\dfrac{6g}{\ell}}$

$\dfrac{f}{f_0} = \sqrt{\dfrac{6g}{\ell}}\sqrt{\dfrac{13\ell}{18g}} = \sqrt{\dfrac{13}{3}}$

17. $K = \frac{1}{2}mv^2$

$\frac{dK}{dt} = \frac{1}{2}v^2\frac{dm}{dt} + \frac{1}{2}m \times 2v\frac{dv}{dt}$

$= \frac{1}{2}9t^2 \times 18 + \frac{1}{2} \times 54 \times 2 \times 3t \times 3$

$\left.\frac{dK}{dt}\right|_{t=2} = 9 \times 2 \times 18 + 54 \times 18$

= 1296 J/s

18. for block B

$m_2g - T = m_2a_2$ …(i)

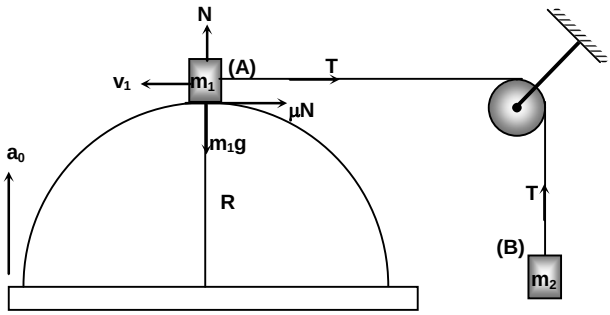
$T + \mu N = m_1a_2$ …(ii)

by circular motion dynamics

$m_1g + m_1a_0 - N = \frac{m_1v_{rel}^2}{R}$ (from hemisphere)

on solving these equation and putting of all variable we get $a_A = 2\sqrt{17}$ m/s^2

and $a_B = 8$ m/s^2



19. $F = u\frac{dm}{dt}$ and $P = Fv$

20. $i = \frac{\varepsilon_0 aV}{d}(k-1)A\omega \sin \omega t$

$q = \int |i|\, dt$

SECTION – D

21. $\Delta\ell_T = \alpha\ell\Delta T$ Extension …(i)

$\frac{mg}{S} = \frac{F}{S} = Y\frac{\Delta\ell_e}{\ell}$

$\Rightarrow \quad \Delta\ell_e = \frac{mg\ell}{SY} \Rightarrow$ compression …(ii)

If there is no change in length, it means extension due to temperature raise must be equal to the elastic compression due to weight. So

$\alpha\ell\Delta T = \frac{mg\ell}{SY}$

$\Rightarrow m = \frac{\alpha SY\Delta T}{g} = \frac{1.1 \times 10^{-5} \times 10 \times 10^{-4} \times 10 \times 2.0 \times 10^{11}}{10} = 2.2 \times 10^3 kg$

22. $T = \frac{2u\sin\theta}{g} = 4$ s

First collision occur after 5/20 sec

So, $\Delta t = 4 - \frac{5}{20} = 3.75$ s

$S_1 = 3.75 \times 10 = 37.5$ m

When ball collide with wall B, then total time elapse

$t = 0.25 + 1.5 = 1.75$ s

$\Delta t = 4 - 1.75 = 2.25$ s

$S = 2.25 \times 2.5 = 5.625$ m

23. Initially $V800 = V'1000$

$\Rightarrow V' = 0.8\ V$ (V' is volume inside water)

$\Rightarrow H' = 0.8\ H = 0.8 \times 50$ cm $= 40$ cm

If we displace the block by dy inside water then

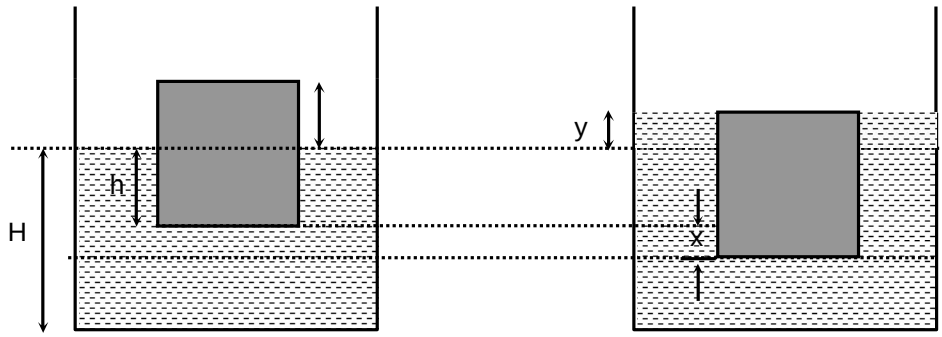
$$dy = dx\left[1 - \frac{a}{A}\right]$$

Where dx is the new dipped part of block on displacing the block by dy

$$\Rightarrow dW = \int_{0.4}^{0.5} \left[x(0.5)^2 1000g - (0.5)^3 800g\right] dx \frac{3}{4} \Rightarrow W = 9.375 \text{ J}$$

Second Method:

a $\Rightarrow$ Area of cube A $\Rightarrow$ Base area of container.



$$AH - ah = A(H + y) - a(h + x + y)$$

$$\Rightarrow \quad y = \frac{a}{A - a}x \Rightarrow x + y = \frac{A}{A - a}x$$

$$F_{ex} = F_B - mg = \rho g a(h + x + y) - a^3 \rho_B g$$

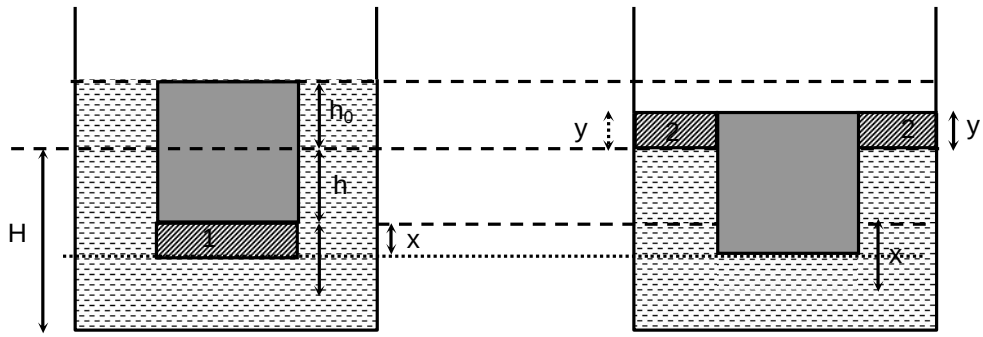
$$= a(x + y)\rho_\ell g = a\left(\frac{A}{A - a}\right)x\rho_\ell g$$

$$W = \int_0^{x_0} F_{ex}\,dx = \frac{aA\rho_\ell g}{(A - a)} \times \frac{x_0^2}{2} \text{ Here } x_0 = \frac{0.2\sqrt{a}(A - a)}{A}$$

$$\Rightarrow W = \frac{aA\rho_\ell g}{(A - a)} \times \frac{a^2}{2} \times \frac{0.04 \times a(A - a)^2}{A^2} = \frac{0.04a^2(A - a)}{2A}\rho_\ell g = \frac{0.04 \times \frac{1}{4} \times \frac{3}{4} \times \frac{1}{4} \times 1000 \times 10}{2 \times 1}$$

$$= \frac{300}{32} = \frac{75}{8} = 9.375 \text{ Joule}$$

Third Method: $a = \ell \times \ell$



$h_0 = \frac{\ell}{5},\ x + y = \frac{\ell}{5},\ x = 3y \Rightarrow x = \frac{3\ell}{20}$

$m_0 \Rightarrow$ mass of liquid displace = $ax\rho_\ell = \frac{3a\rho_\ell}{20} = \frac{3V\rho_\ell}{20}$

Vertical displacement of liquid = $h + \frac{x+y}{2} = \frac{4\ell}{5} + \frac{\ell}{10} = \frac{9\ell}{10}$

$W_{FB} \Rightarrow$ work done by force of buoyancy to raise the liquid $= -m_0 g\frac{g\ell}{10} = -\frac{3V}{20}\rho_\ell \times g \times \frac{9\ell}{10}$

$= -\frac{3}{20} \times \frac{1}{8} \times 1000 \times 10 \times \frac{9}{10} \times \frac{1}{2} = -84.375$ J

$W_{mg} \Rightarrow$ Work done by weight to displace the body downward by distance x

$= mgx = \frac{120}{8} \times 800 \times 10 \times \frac{3}{20} \times \frac{1}{2} = 75$ J

According work energy theorem

$W_{Fex} + W_{FB} + W_{mg} = 0$

$\Rightarrow W_{Fex} = -W_{FB} - W_{mg} = 84.375 - 75 = 9.375$ Joule

Chemistry

PART – II

SECTION – A

24. $B > Tl > Ga > Al > In$

kJ / mole : 800 589 579 577 558

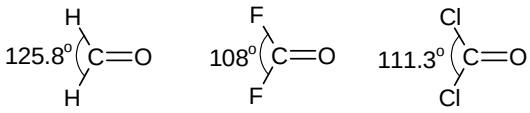
The last element (Tl) has exceptionally high first ionisation energy due to poor screening caused by f-electrons.
The second element (Al) has exceptionally low first ionisation energy due to good screening caused by s & p electrons.

$$HCOOH + HCOOH \rightleftharpoons HCOOH_2^+ + HCOO^-$$

25. 25M 25M 0 0

$25 - 10^{-3}$ 10^{-3} 10^{-3}

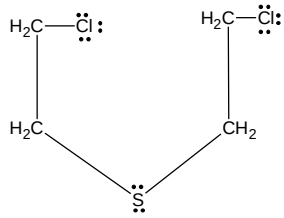
$$\% = \frac{10^{-3}}{25} \times 100 = 0.004\%$$

26. 

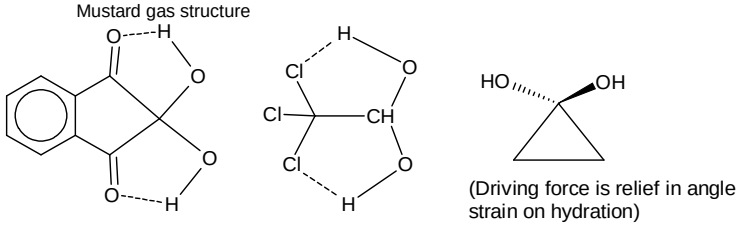
Higher the electronegativity of bonded atom, lesser is the bond angle between the bonds.

27. 

$$Cl_2O + H_2O \longrightarrow 2HOCl$$

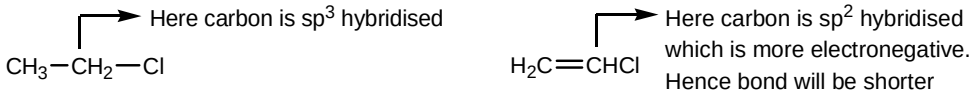
28. 

Mustard gas structure

29. 

(Driving force is relief in angle strain on hydration)

30. The formula of complex is:
$[CrCl_2(en)_4]_2\,[PdCl_4]$
Hence, *i = 3.*

31. (A) 

(B) $CH_2=CH_2$ — 1.34 Å — No resonance

$CH_2=CH-Cl$ — 1.38 Å — In vinylchloride C=C bond length is longer due to lone pair and double bond resonance

(C) C–C bond length are different in all three cases

(D) CH_3-CH_2-Cl — 2.05 D

$CH_2 = CHCl$ — 1.44 D

In $CH_2 = CH - Cl$ charge separation is less.

32. E.W
$N_2 = 42$
$H_3PO_2 = 33$
$HN_3 = 43$
$P_4O_6 = 27.5$

33. $12N_2 + 4H_3PO_2 \longrightarrow 8HN_3 + (P_4O_6 + 2H_2O)$

$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$

6 moles H_2O reacts with 1 mole P_4O_6

2 moles H_2O reacts with $\frac{1\times 2}{6} = \frac{1}{3}$ mole P_4O_6

$\therefore$ fraction of P_4O_6 left unreacted = $\frac{2}{3}$

SECTION – C

34. pH range of indicator = 3.4 – 4.6
$\therefore$ pOH range of indicator = 9.4 – 10.6

$\therefore pK_b$ of indicator $= \frac{9.4+10.6}{2} = 10$

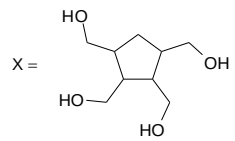
$InOH \rightleftharpoons In^+ + OH^-$

$pOH = pK_b + \log\frac{[In^+]}{[InOH]}$

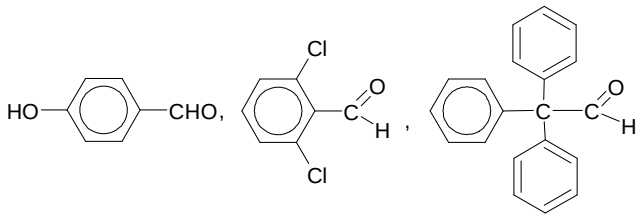
For solution to appear in the colour of In^+, $pOH \geq 10.6$

In limiting case; $10.6 = 10 + \log\frac{[In^+]}{[InOH]}$ $\quad \therefore \frac{[In^+]}{[InOH]} = 4$

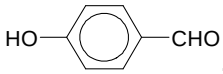
35. X =



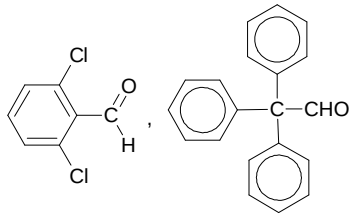
36.



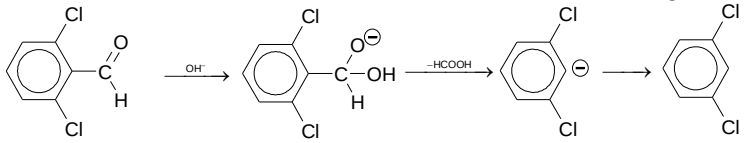
Do not contain any α-hydrogen but still fails to undergo Cannizaro reaction. In case of



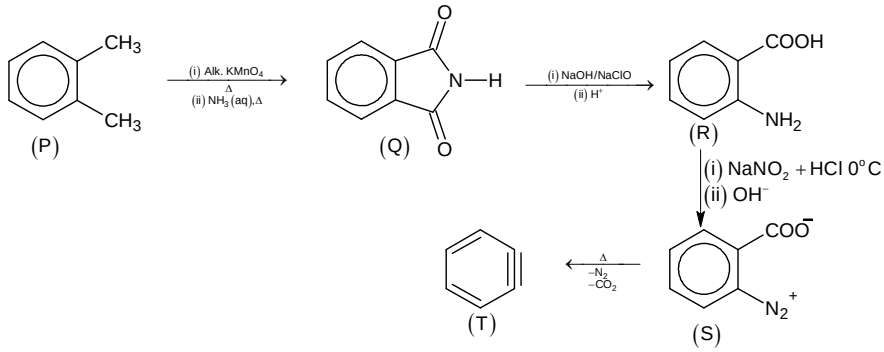
the corresponding hydroxide is strongly stabilised by –R effect of –CHO. Consequently, the electrophilicity of the carbonyl carbon is reduced and the OH^- does not attack the carbonyl group.

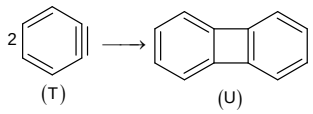
In case of 

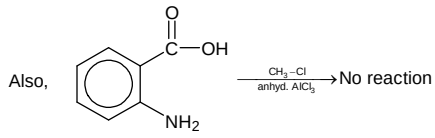
The intermediate formed due to attack of OH^- cleaves to give a stable anion.



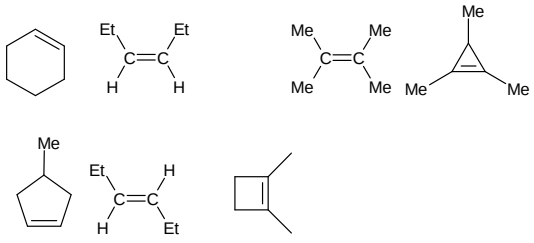
37.







38.



39. $C_3H_8O_3 + nCH_3MgI \longrightarrow nCH_4$

mass 0.092g

moles $\frac{0.92}{92}$mole $\quad\quad$ $\frac{67}{22400}$mole

$=1$m mole $\quad\quad$ $= 3$ m mole

So, on comparing each molecule of $C_3H_8O_3$ must give 3 molecule of CH_4

40. $\text{Rate} = k\left[C_3H_6^*\right]$ (i)

$$Kc(\text{for step1}) = \frac{[C_3H_6]\left[C_3H_6^*\right]}{[C_3H_6][C_3H_6]} = \frac{\left[C_3H_6^*\right]}{[C_3H_6]}$$

$$\therefore \left[C_3H_6^*\right] = K_c\left[C_3H_6\right]$$

Substituting in equation (i)

$R = k.\ K_c[C_3H_6]$

$R = k'[C_3H_6]$

41. $\nu = \frac{v}{2\pi r}$

$v \propto \frac{Z}{n}$

$r \propto \frac{n^2}{Z}$

$\nu \propto \frac{Z^2}{n^3}$

$$\frac{\nu_{He^+}}{\nu_H} = \frac{4}{27} \div \frac{1}{8} = \frac{32}{27} = \frac{2^5}{3^3}$$

$x = 5$

$y = 3$

$x + y = 8$

42. b, c, d, f

43. One mole of the above compound (X) on hydrolysis followed by heating produces 4 moles of CO_2.

Hence, 0.5 M of X produce 2 moles of CO_2.

SECTION – D

44. $Ca_3(PO_4)_2 + 8Mg \longrightarrow \underset{(A)}{Ca_3P_2} + \underset{(B)}{8MgO}$

$\underset{(A)}{Ca_3P_2} + 6H_2O \longrightarrow 3Ca(OH)_2 + \underset{(C)}{2PH_3}$

$2PH_3 + 3CuSO_4 \longrightarrow \underset{(X,\text{Black})}{Cu_3P_2}\downarrow + 3H_2SO_4$

46. $\rightarrow v = \left(\frac{2\pi Ke^2}{h}\right) \times \frac{Z}{n}$, so, atomic no. (Z) $\propto$ slope of the graph.

So, statement (1) is True.

$\rightarrow \lambda = \frac{h}{\sqrt{2qVm}}$, slope of the graph $\propto \frac{1}{\sqrt{m}}$

So, statement (2) is True.

$\rightarrow$From, PV = nRT

$V = \left(\frac{nR}{P}\right)T$, so, slope of graph $\propto \frac{1}{P}$

So, statement (3) is also correct.

$\rightarrow P = (nRT)\frac{1}{V}$, so, slope $\propto$ T

So, statement (4) is incorrect.

Mathematics

PART – III

SECTION – A

47. $\frac{1}{3}\frac{d}{dx}\left\{e^{x^3}\left(y^3-1\right)\right\}+\frac{1}{3}\frac{d}{dx}\left(e^{y^3}\right)=0$

48. Each prime can go in one of the five slots – x_1, x_2, x_3, x_4 and none of these.

49. $4^4\left(\frac{\sin^{12}B+\sin^{12}C}{\left(\sin 2A+\sin 2B+\sin 2C\right)^4}\right) = \frac{\alpha^4+\beta^4}{\left(\frac{\alpha+\beta}{2}\right)^4}$ where α, β are distinct positive real number whose AM is a and two GM's are b, c

50. Any combination of 4 points gives 3 new intersection points

51. $f(1-x)=f(1+x)$
$\Rightarrow f'(1+x)=-f'(1-x)$

52. $I<\int_0^{\pi/2}\frac{\cos\theta}{10-6\cos\theta}d\theta<\frac{1}{4}\int_0^{\pi/2}\cos\theta d\theta<1$
$\frac{\cos\theta}{10-6\cos\theta}>\left|\frac{\cos(\pi-\theta)}{10-6\cos(\pi-\theta)}\right|$ for $\theta\in\left[0,\frac{\pi}{2}\right)\Rightarrow I>0$

53. For area of parallelogram PQRS to be least line L makes angle $\frac{\pi}{4}$ with L_1 and L_2

54. Centre of the circle is at (a, a)
$a^2+(a-2)^2=(a-4)^2+(a-3)^2$
$\Rightarrow a=\frac{21}{10}$

55. D_5

56. ${}^5C_1\left\{{}^4C_1\cdot D_4+{}^4C_1\cdot\left({}^3C_1\cdot D_3+{}^3C_1\times{}^2C_1\right)\right\}$

SECTION – C

57. ${}^3C_1 \times D_3 + {}^3C_1 \times D_2 \times {}^2C_1$

58. Let $x = r\cos\theta$; $y = r\sin\theta$

$\Rightarrow r^2 = \dfrac{\sin 2\theta}{\cos^4\theta + \sin^4\theta}$

59. Exactly four girls are together – Exactly four girls are together and B_1 B_2 are together

$p = \dfrac{5! \times 8! \times 58}{13!}$

60. $\tan\alpha = \dfrac{3}{5}$, $\tan\beta = \dfrac{5}{4\cos\alpha} = \dfrac{\sqrt{34}}{4}$

61. $9\left(1 - \dfrac{1}{3}\right) = 9 \times \dfrac{2}{3} = 6$

62. $\Delta = \begin{vmatrix} \gamma^2 & \beta^2 & 2\beta\gamma - \alpha^2 \\ 2\alpha\gamma - \beta^2 & \alpha^2 & \gamma^2 \\ \alpha^2 & 2\alpha\beta - \gamma^2 & \beta^2 \end{vmatrix} = \left(\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma\right)^2$

63. Denominator = ${}^nC_0\ {}^{n+k-1}C_{n-1} + {}^nC_1\ {}^{n+k-2}C_{n-1} + {}^nC_2\ {}^{n+k-3}C_{n-1} + \ldots {}^nC_{k-1}\ {}^nC_{n-1} + {}^nC_k\ {}^{n-1}C_{n-1}$

= coefficient of x^{n-1} in

${}^nC_0 (1+x)^{n+k-1} + {}^nC_1 (1+x)^{n+k-2} + {}^nC_2 (1+x)^{n+k-3} + \ldots {}^nC_{k-1} (1+x)^n + {}^nC_k (1+x)^{n-1}$

= coefficient of x^{n-1} in

${}^nC_0 (1+x)^{n+k-1} + {}^nC_1 (1+x)^{n+k-2} + {}^nC_2 (1+x)^{n+k-3} + \ldots {}^nC_{k-1} (1+x)^n + {}^nC_k (1+x)^{n-1}$
$+ {}^nC_{k+1} (1+x)^{n-2} + {}^nC_{k+2} (1+x)^{n-3} + \ldots {}^nC_n (1+x)^{k-1}$

= coefficient of x^{n-1}

$(1+x)^{n+k-1}\left(1 + (1+x)^{-1}\right)^n = (1+x)^{k-1}(x+2)^n$

= coefficient of x^{n-1} in

$\sum_{r=1}^{n} {}^nC_r\ 2^r\ x^{n-r}(1+x)^{k-1} = \sum_{r=1}^{n} 2^r\ {}^nC_r\ {}^{k-r}C_{r-1}$.

64. Let r_1, r_2 and r are in-radius of ΔABD, ΔADC and ΔABC respectively

$r_1 = \dfrac{c}{a} \times r$; $r_2 = \dfrac{b}{a} \times r$ and $EF = \sqrt{2\left(r_1^2 + r_2^2\right)} = \sqrt{2}\,r = AI$

65. $\sum_{r=1}^{m} \dfrac{(m+1)(r-1)m^{r-1}}{r\ {}^mC_r} = \sum_{r=1}^{m} \left(\dfrac{\left[mr - (m-r+1)\right]m^{r-1}}{r\ {}^mC_r} \right)$

$$= \sum_{r=1}^{m}\left(\frac{m^r}{{}^mC_r} - \frac{m^{r-1}(m-r+1)\,r!\,(m-r)!}{r.\ m!}\right)$$

$$= \sum_{r=1}^{m}\left(\frac{m^r}{{}^mC_r} - \frac{m^{r-1}(m-r+1)\,!\,(r-1)!}{m!}\right) = \sum_{r=1}^{m}\left(\frac{m^r}{{}^mC_r} - \frac{m^{r-1}}{{}^mC_{r-1}}\right) = m^m - 1.$$

66. Let the variable chord be $x\cos\alpha + y\sin\alpha = p$. Let this chord intersect the hyperbola in A and B

The joint equation of OA and OB is $\frac{x^2}{4} - \frac{y^2}{8} = \left(\frac{x\cos\alpha + y\sin\alpha}{p}\right)^2$

$$\Rightarrow \left(\frac{1}{4} - \frac{\cos^2\alpha}{p^2}\right)x^2 - \left(\frac{1}{8} + \frac{\sin^2\alpha}{p^2}\right)y^2 - \frac{2\sin\alpha\cos\alpha}{p^2}xy = 0$$

$$\Rightarrow \frac{1}{4} - \frac{\cos^2\alpha}{p^2} - \frac{1}{8} - \frac{\sin^2\alpha}{p^2} = 0 \Rightarrow p^2 = 8.$$

The variable line touches the fixed circle, thus perpendicular distance of (0, 0) = Radius

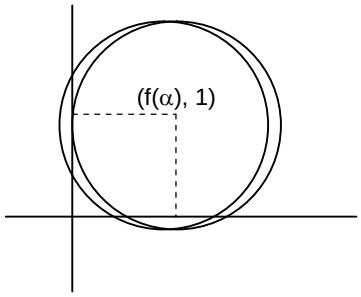
$\therefore$ Equation of the circle is $x^2 + y^2 = 8$

SECTION – D

67. $p = \frac{{}^5C_1}{{}^7C_3}\cdot\frac{2}{3}\cdot\left(\frac{1}{2}\right)^4$

68. L = circumference of unit circle = 2π

69. From the figure it is clear that for $\arg z \geq \frac{\pi}{2}$ circle should either lie in second quadrant or touch the positive imaginary axis.



$\Rightarrow f(\alpha) \leq 1$

$\alpha^2 - 7\alpha + 13 \leq 1$

$\alpha^2 - 7\alpha + 12 \leq 0$

$3 \leq \alpha \leq 4$